

Questions			Answers	Marks
8	(a)		beta (minus) decay	B1
			<u>charge / proton number need to be conserved</u> so with increase in proton number <u>and</u> a particle of negative charge must be emitted. Alternative Also accept nuclear equation	B1
	(b)	(i)	Use to kill cancer cells (by implanting it near the tissues)	M1
			half-life is short so duration of powerful radiation will not be too long or beta particles do not travel far in tissue, so surrounding cell will not be damaged.	A1
		(ii)	1. Using $A = \lambda N$, $64\,000 = \frac{\ln 2}{2.69 \times 24 \times 60 \times 60} N$ to get $N = 2.14(60)$	B1
			$N = \frac{6.02 \times 10^{23}}{198} \times \text{mass in gram}$ $m = 7.06 \times 10^{-12} \text{ g}$	A1
			2. $A = 64\,000 e^{-\frac{\ln 2}{2.69} \times 13.5} = 64\,000 \times 0.03085$	M1
			$A = 1.97 \text{ kBq}$	A1

Section B

Questions	Answers	Mar
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